

Bit Manipulations

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Introduction

In this lab you will learn to use different techniques for the manipulation of individual bits in registers. You will use these to read dip switches and buttons, as well as to write out values. Specifically, you will detect events on the four buttons `T[3..0]`.

Learning Objectives

- You can apply the bitwise operators in C for setting and clearing individual register bits.
- You can explain the purpose of an edge detection and are able to implement it in C.
- You know how to use the bit shifting operators in C.

Task 1: Control individual LEDs

Expand the given program so that `LED7..6` are always on (bright) and `LED[5..4]` are always off (dark). As a result `LED[3..0]` will be controlled by the settings of DIP switches `SW[3..0]` whereas the state of the four leftmost DIP switches will be ignored.

Define (`#define`) a mask for the “bright” as well as for the “dark” bits and use the assignment operators `|=` and `&=` to set/clear the corresponding bits.

Verify the correct behaviour with different positions of the DIP switches.

Task 2: Edge detection

- Expand the program from [Task 1](#). Read the state of the buttons `T[3..0]` into a variable, that you define. Mask the unused upper four bits. Define an 8 bit wide counter variable. Increment this variable each time the button `T[0]` is pressed i.e. whenever bit 0 is high. Output the counter variable on `LED[15..8]` during each loop iteration.
- What do you notice when testing? By which value does the counter increase when you press the button? What is the cause?
- Expand the program with another counter variable that counts the “push events” of the button. Count the push events independent of how long the button is pressed by the user. Output the variable on `LED[31..24]`. Implement an edge detection, which detects if one or more buttons were not pressed (low) during the last loop iteration and are now pressed (high).

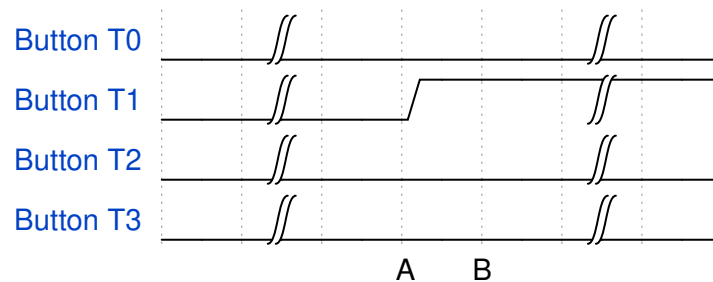


Figure 1: edge detection

A	B	Z
0	0	0
0	1	1
1	0	0
1	1	0

- At the end of the loop save the value of the button state in a variable.
- Implement the edge detection for all four buttons at the same time. You can do this by comparing the last button state with the current. Use the bitwise NOT ($\sim$) and the AND ($\&$), or the bitwise XOR ($\wedge$).
- Then use a bit mask to test for button T0.

Task 3: Add functions for remaining buttons

Expand your program further. Implement the following functions for the buttons. Use a variable of type `uint8_t`.

- **Button T3** - The variable shall be assigned to value set on the DIP switches `SW[7..0]`
- **Button T2** - The value of the variable shall be inverted bitwise
- **Button T1** - The value of the variable shall be shifted to the left by one bit ($\ll$)
- **Button T0** - The value of the variable shall be shifted to the right by one bit ($\gg$)

In the case where more than one button is pressed simultaneously only one shall be carried out. Continuously output the value of the variable on `LED[23..16]`.

Task 4 (optional)

Change the program from [Task 3](#), so that only bits `[5..2]` are inverted when button `T[2]` is pressed. The other bits shall be left untouched.

You can toggle a bit with the XOR operation ($\wedge$). Create a mask with the bits to toggle set (true).

I/O Overview of combined programs

In the following depiction you see all in- and outputs of today's tasks.

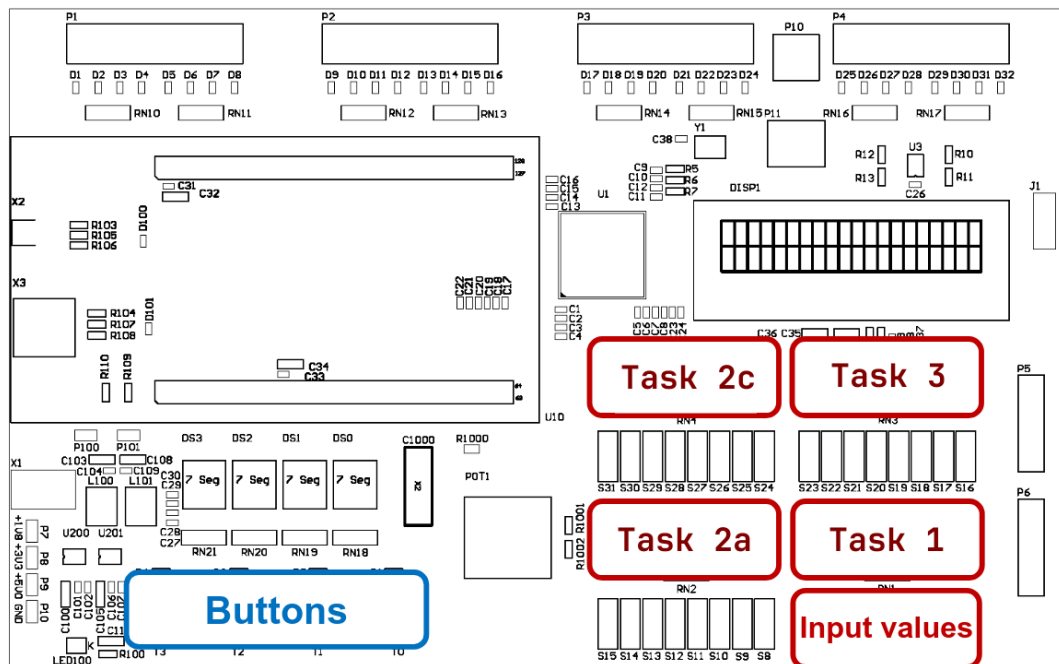


Figure 2: In- and output

Grading

The working programs have to be presented to the lecturer. The student has to understand the solution / source code and has to be able to explain it to the lecturer.

Criteria	Weight
Task 1 implemented	1 / 4
Task 2 a) implemented, b) explained	1 / 4
Task 2 c) implemented	1 / 4
Task 3 implemented	1 / 4